

Carlos Roberto de Melo Carneiro

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Postdoctoral Researcher in Extragalactic Astrophysics at Shanghai Jiao Tong University focusing on strong gravitational lensing and stellar dynamics. My work centres on joint lensing–dynamics modelling to constrain the mass distributions of galaxies, providing empirical benchmarks to probe dark matter, modified gravity, and galaxy evolution. More recently, my research extends to scaling relations between supermassive black holes and their host galaxy properties.

RESEARCH INTERESTS

My research interests span strong gravitational lensing, stellar dynamics, galaxy evolution, and cosmology. I am particularly focused on the joint modelling of strong gravitational lensing and stellar dynamics to investigate the mass distribution of galaxies and its connection to dark matter, galaxy evolution, and gravity. I am also interested in developing new methodologies to improve these analyses, including advanced statistical techniques, hydrodynamical simulations, and novel dynamical tracers.

RESEARCH EXPERIENCE

Shanghai Jiao Tong University (SJTU)
Postdoctoral Fellow
Research on strong gravitational lens modelling and black hole scaling relations

Shanghai, China
January/2026 -

EDUCATION

Universidade Federal do Rio Grande do Sul (UFRGS)
PhD in Astrophysics
Tracing mass in the universe, one galaxy at a time:
A unified lensing and dynamics approach to probe gravity and galaxy evolution
Advisors: Cristina Furlanetto & Ana Chies Santos

Porto Alegre, Brazil
August/2021 - August/2025

Institute of Cosmology and Gravitation (ICG)
PhD internship in Astrophysics
Advisor: Thomas Collett

Portsmouth, UK
April/2024 - September/2024

Universidade Federal do Rio Grande do Sul (UFRGS)
MSc in Astrophysics
Testing General Relativity at Galactic Scales
Advisors: Cristina Furlanetto & Ana Chies Santos

Porto Alegre, Brazil
August/2019 - August/2021

Universidade Federal de Juiz de Fora (UFJF)
BSc in Physics
Cosmological Models of accelerated expansion: Structure Formation and Event Horizon
Advisor: Gil de Oliveira Neto

Juiz de Fora, Brazil
March/2015 - July/2019

ACADEMIC HONOURS

2021-2025	PhD research fellowship granted by CNPq
2024/04- 2024/10	PDSE-Print international research fellowship granted by CAPES
2019-2021	Master research fellowship granted by CNPq
2018-2019	Undergraduate research fellowship granted by UFJF
2016-2017	Undergraduate research fellowship granted by FAPEMIG

PUBLICATIONS

1. **Melo-Carneiro C. R.**, et al., *Unveiling a 36 Billion Solar Mass Black Hole at the Centre of the Cosmic Horseshoe Gravitational Lens*, [MNRAS \(2025\), 541, 2853](#)
2. Li, Sung Kei, ..., **Melo-Carneiro C. R.**, et al., *Lensed stars in galaxy-galaxy strong lensing - a prediction for the Cosmic Horseshoe*, [MNRAS \(2025\), 544, 4469](#)
3. R. A. Riffel, **Melo-Carneiro C. R.**, et al., *Blowing star formation away in AGN hosts (BAH) III. Serendipitous discovery of a $z \sim 2.9$ star-forming galaxy lensed by the galactic bulge of CGCG 012-070 using JWST NIRSpec*, [A&A \(2025\), 701, L11](#)
4. **Melo-Carneiro C. R.**, Furlanetto C., Chies-Santos A. L., *Systematics in ETG Mass Profile Modelling: Strong Lensing & Stellar Dynamics*, [Journal of Cosmology and Astroparticle Physics, Volume 2025, January 2025, 2025](#)
5. **Melo-Carneiro C. R.**, Furlanetto C., Chies-Santos A. L., *Probing general relativity in galactic scales at $z \sim 0.3$* , [MNRAS \(2023\), 520, 1613](#).
6. **Melo-Carneiro C. R.**, Furlanetto C., Chies-Santos A. L., *Constraining general relativity at $z \sim 0.3$: MUSE Kinematics of SDP.81*, [Proceedings of the International Astronomical Union 359: Galaxy Evolution and Feedback across Different Environments \(2019\), pp. 260–261](#).

SUCCESSFUL PROPOSALS

- **Testing General Relativity in Galaxy Scales**, April 2021–April 2023. Allocation time for the supercomputer Santos Dumont ([SDumont](#)) at the National Laboratory of Scientific Computation (Laboratório Nacional de Computação Científica - [LNCC](#)). Our proposal was accepted for the Standard program with a total of 2073600 AUs. This program was extended for one more year until April 2024.
- **Confirming strongly lensed galaxies with the SOAR Integral Field Spectrograph - 40h**, 2023 (Co-I). The proposal aimed to confirm the presence of strong gravitational lensing in 12 galaxy-galaxy candidate systems and three quadruple quasar candidates. Observations were conducted using the SOAR Integral Field Spectrograph (SIFS).
- **Testing General Relativity in Galaxy Scales**, April 2024–Current. Allocation time for the supercomputer Santos Dumont ([SDumont](#)) at the National Laboratory of Scientific Computation (Laboratório Nacional de Computação Científica - [LNCC](#)). Our proposal was accepted for the Standard program with a total of 2073600 AUs. This accepted proposal is a continuation of the previous project.
- **COHORTS: COsmic HORseshoe Transient Survey – observing lensed stars at the cosmic noon under galaxy–galaxy strong lensing -16.3 h**, JWST Cycle 5 (Co-I; to be observed in 2027). Targets the Cosmic Horseshoe ($z = 2.4$) with two NIRCcam visits spaced by 30 days to perform the first search for microlensed stars in a galaxy-scale strong lens.

SCIENTIFIC TALKS and POSTERS

- 2026 **Precision Cosmology with Strong Lensing Time Delays - Tokyo, Japan**
Talk. Title: Stress-Testing Lensing-Only Slope Measurements
- 2025 **GalaxyMemoirs: Inferring their past from their present - Búzios, Brazil**
Poster. Title: Unveiling a 36 Billion Solar Mass Black Hole at the Heart of the Cosmic Horseshoe Gravitational Lens
- 2025 **COSMO Seminar at Brazilian Center for Research in Physics (CBPF) - Rio de Janeiro, Brazil**
Oral. Title: Unveiling a 36 Billion Solar Mass Black Hole at the Heart of the Cosmic Horseshoe Gravitational Lens
- 2024 **Measures of Luminous and Dark Matter in Galaxies Across Time at XXXII IAU General Assembly - Cape Town, South Africa**
Oral. Title: Self-consistent Modelling of Strong Gravitational Lensing and Stellar Dynamics
- 2024 **Strong Gravitational Lensing Science with LSST - Virtual Participation**
Poster. Title: Self-consistent Modelling of Strong Gravitational Lensing and Stellar Dynamics
- 2023 **European Astronomical Society Annual Meeting - Virtual Participation**
Poster. Title: Probing General Relativity in galactic scales at $z \sim 0.3$
- 2023 **IAUS 381: Strong Gravitational Lensing in the Era of Big Data - Otranto, Italy**
Poster. Title: Probing General Relativity in galactic scales at $z \sim 0.3$
- 2022 **25th annual International Conference on Particle Physics and Cosmology (COSMO'22) - Rio de Janeiro, Brazil**
Oral. Title: Does General Relativity hold in galactic scales? A test at a $z \sim 0.3$ elliptical lens galaxy.
- 2022 **XLV Brazilian Astronomical Society virtual meeting - Virtual Meeting**
Oral. Title: Does General Relativity hold in galactic scales? A test at a $z \sim 0.3$ elliptical lens galaxy.
- 2020 **IAUS 359: Galaxy Evolution and Feedback Across Different Environments (GALFEED) - Bento Gonçalves, Brazil**
Poster. Title: Constraining General Relativity at $z < 0.3$

SCHOOLS and WORKSHOPS

- 2026 **Precision Cosmology with Strong Lensing Time Delays**
Organised by the Research Center for the Early Universe (RESCEU, University of Tokyo). The workshop focused on recent progress and future prospects in lens modelling, lensed supernovae, time-delay cosmography, substructure studies, and synergies with current and upcoming facilities.
- 2026 **Montpellier Strong Gravitational Lensing Workshop 2026**
Hosted by the Laboratoire Univers et Particules de Montpellier (Université de Montpellier). Topics covered cutting-edge developments in strong lensing: challenges in lens-mass modelling; line-of-sight corrections; multi-plane and compound lens systems; sub-halo and line-of-sight halo searches; and wave-optics effects.
- 2024 **III Workshop on Lensing by Galaxies and Clusters**
Workshop held by the Brazilian Center for Research in Physics (CBPF, Brazil). Workshop on weak and strong lensing studies on galaxy and galaxy cluster scales; applications to cosmology and modified gravity tests.

- 2022 **DYNAMITE (DYnamics, Age and Metallicity Indicators Tracing Evolution)**
Workshop held by University of Western Australia (UWA, AUS). Workshop on triaxial Schwarzschild orbit-based dynamical models.
- 2022 **Summer School in Statistics for Astronomers XVII**
School held by Pennsylvania State University (PSU, US). Summer school on astrostatistics.
- 2020 **Partículas, AStropartículas, Campos e COsmologia - PASCCO**
School held by International Institute of Physics (IIP, Brazil). School of Particles, Astroparticles, Fields and Cosmology.
- 2020 **JWST-UFRGS Workshop**
Workshop held by Universidade Federal do Rio Grande do Sul (UFRGS, Brazil). JWST writing proposal workshop.

TEACHING EXPERIENCE

- 2020 Teaching assistant at undergraduate Astrophysics course: Fundamentals of Astronomy and Astrophysics, UFRGS
- 2018 Tutor at undergraduate Physics course: Physics III - Electromagnetism (UFJF)
- 2017 Tutor at undergraduate Mathematics course: Calculus II (UFJF)

SKILLS and LANGUAGES

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| Computer Skills | Python, LaTeX, SLURM/PBS |
| Astronomy tools | DS9, QFitsView, pXF, MgeFit, JamPy, PyAutoLens, MPDAF, Astropy |
| Other useful tools | SciPy, NumPy, dynesty, emcee, Jupyter, VSCode |
| Languages | Proficient in Portuguese and English. |

REVIEWING

Reviews for MNRAS and Nature Astronomy were performed during my PhD under the supervision of my advisor as part of a referee training programme.

- *Monthly Notices of the Royal Astronomical Society*
- *Nature Astronomy*
- *Nature Communications*
- *Cadernos de Astronomia*

OUTREACH & MEDIA

- *'Most massive black hole ever discovered' is detected*, [Royal Astronomical Society](#), 07 August 2025.
- *New method reveals perhaps the most massive black hole yet spotted*, [AAAS | Science](#), 11 August 2025.
- *Pesquisa detecta buraco negro gigante, 36 bilhões de vezes a massa do Sol*, [UFRGS](#), 07 August 2025.
- *Brasileiro descobre buraco negro 36 bilhões de vezes mais pesado que o Sol*, [Revista Galileu](#), 07 August 2025.
- *AstroUFRGS Podcast (2021–2022)*: outreach podcast of the UFRGS Astronomy Department, where I participated as host and speaker, [Spotify](#).